

EEE 460 (July 2025)

Optoelectronics Laboratory

Final Project Report

Section: G2 Group: 06

Design of a Solar Curve Tracer Using SEPIC Converter

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1 Abstract:

This project presents the design, simulation, and hardware implementation of a low-cost photovoltaic (PV) I–V curve tracer using a SEPIC (Single-Ended Primary Inductor Converter) topology. Accurate characterization of PV modules requires full I–V and P–V curve extraction under varying operating conditions; however, conventional methods such as resistive and capacitive loading are often slow, bulky, and less suitable for real-time or higher-power applications. To overcome these limitations, a DC–DC converter-based approach is adopted, where the SEPIC converter acts as a controllable electronic load.

By varying the duty cycle of the SEPIC converter, the equivalent input resistance seen by the PV module is dynamically adjusted, enabling smooth traversal of the operating point from near short-circuit current to near open-circuit voltage. At each duty-cycle step, the PV voltage and current are measured using an Arduino Uno-based data acquisition system integrated with INA219 current/voltage sensor and BH1750 light sensor for irradiance estimation. The collected data is processed to generate I–V and P–V characteristics, providing insight into the performance of the solar module.

The system was first modeled and verified in MATLAB/Simulink to analyze converter behavior and ensure correct tracing methodology. A hardware prototype was then implemented on a custom-designed PCB and tested using low-power PV modules (1.5 W and 15 W). Experimental results demonstrate that the proposed system can accurately capture I–V and P–V curves and track key parameters such as maximum power point (MPP).

The proposed design offers several advantages, including fast and continuous curve tracing, compact hardware implementation, low cost, and suitability for educational and laboratory use. Additionally, the integration of sensing, control, and data processing into a single platform makes the system practical for real-time PV performance analysis. This work highlights the effectiveness of SEPIC-based converters in PV characterization and provides a scalable foundation for future improvements such as higher-power capability, improved accuracy, and wireless data monitoring.

2 Introduction

Photovoltaic (PV) energy has become one of the most important renewable energy sources because of its clean operation, wide availability, and growing role in sustainable power generation. The electrical behavior of a PV module is commonly represented by its current–voltage (I–V) and power–voltage (P–V) characteristics, which provide important information about short-circuit current, open-circuit voltage, maximum power point, and overall panel performance. Since these characteristics vary with irradiance, temperature, and load condition, accurate curve tracing is necessary for proper performance evaluation, comparison with datasheet values, and detection of abnormal operating conditions. Therefore, the ability to measure I–V and P–V curves in real operating conditions is highly valuable in photovoltaic analysis, testing, and educational laboratory work.

Several methods have been proposed for obtaining PV curves, including variable resistive load, capacitive load, electronic load, and DC–DC converter-based approaches. Conventional methods often suffer from limitations such as low speed, bulky hardware, large power loss,

or difficulty in achieving smooth full-curve scanning. In contrast, converter-based tracing methods use a power electronic converter as a controllable load, allowing the equivalent resistance seen by the PV panel to be changed through duty-cycle variation. This makes it possible to sweep the operating point from near short-circuit condition to near open-circuit condition in a faster and more flexible way. Among the available converter topologies, SEPIC is a suitable choice because it offers buck–boost operation without output polarity inversion and supports practical hardware implementation. Based on this principle, this project presents the design, simulation, and hardware implementation of a low-cost solar curve tracer using a SEPIC converter for small PV modules. The system is verified in MATLAB/Simulink and implemented using an Arduino-based sensing and control platform to generate practical I-V and P-V characteristics for laboratory-scale solar panel analysis.

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

The scope of this project is the development of a photovoltaic curve tracer capable of obtaining the current–voltage (I–V) and power–voltage (P–V) characteristics of low-power solar panels using a SEPIC DC–DC converter. The project includes the theoretical study of PV module behavior, selection of a suitable converter topology, design of the control method based on duty-cycle variation, and measurement of electrical parameters during the tracing process. The system is intended to work as a practical laboratory-scale prototype for small PV modules such as 1.5 W and 15 W panels.

The project scope also covers simulation-based verification and hardware realization. MATLAB/Simulink is used to analyze the converter operation and validate the tracing principle before implementation, while a custom PCB-based prototype is developed for practical testing. The hardware system integrates an Arduino Uno for control and data acquisition, INA219 for voltage-current sensing, and BH1750 for light intensity measurement. The measured data are used to plot I–V and P–V curves and to observe the actual operating characteristics of the PV panel under real conditions.

However, the project is limited to low-voltage and low-current operation because of sensor and hardware constraints. It is not intended for direct use with large PV strings or commercial high-power modules. The main purpose is to develop a compact, low-cost, educational, and practically implementable system that demonstrates the working principle of converter-based PV curve tracing.

3.1.2 Literature Review

The electrical performance of a photovoltaic module is commonly evaluated through its I–V and P–V characteristics, since these curves provide important information about open-circuit

voltage, short-circuit current, maximum power point, and overall module condition. Several methods have been reported in the literature for experimental acquisition of these curves. Traditional methods include variable resistive load, capacitive load, electronic load, and DC–DC converter-based techniques. Each method has its own advantages and limitations in terms of speed, complexity, power handling, and suitability for practical implementation. The general review of these methods is discussed in the uploaded paper on PV tracer development, where resistive, capacitive, electronic, and converter-based methods are compared.

The variable resistive-load method is one of the simplest approaches because it directly changes the load connected to the PV module and records the corresponding current and voltage points. Although it is simple and inexpensive, this method is slow, bulky, and less practical for higher-power modules because large power resistors are needed and continuous curve coverage is difficult. Capacitive-load methods improve speed by charging a capacitor directly from the PV module, but they require careful switching, pre-discharge arrangements, and proper capacitor selection. Electronic-load methods provide better controllability by using active devices such as MOSFETs, but the dissipated power and thermal management become significant concerns.

In comparison, DC–DC converter-based tracing methods offer better performance because the converter behaves like a controllable variable resistance whose value changes with duty cycle. This method allows fast scanning of the complete curve and can be implemented more efficiently than purely dissipative methods. Literature shows that buck–boost family converters such as Cuk and SEPIC are suitable for this purpose because they can emulate a wide range of equivalent input resistance values, enabling scanning from near short-circuit to near open-circuit conditions. The uploaded IEEE paper on portable PV curve tracer explains this principle clearly and presents the relation between converter gain and equivalent PV-side resistance.

Among available topologies, SEPIC is attractive because it can operate in both buck and boost regions, does not invert output polarity, and is convenient for practical hardware design. The uploaded WCNPS paper also notes that SEPIC and Cuk are well suited for PV curve tracing because they share buck–boost characteristics, while SEPIC additionally avoids output polarity inversion and supports galvanic isolation benefits. Based on these observations, a SEPIC-based approach is adopted in this project as a practical compromise between accuracy, controllability, simplicity, and hardware feasibility.

3.1.3 Formulation of Problem

Photovoltaic panels do not produce fixed electrical output; their voltage, current, and power vary with solar irradiance, temperature, and load condition. Therefore, accurate characterization of a PV panel requires tracing its full I – V and P – V curves rather than relying only on datasheet values. In practical laboratory and field situations, this creates an engineering problem: a suitable method is needed to vary the load seen by the PV panel smoothly and quickly so that the full curve can be captured under nearly constant environmental conditions.

The problem becomes more challenging when conventional methods are considered. Resistive loading is simple but bulky and inefficient. Capacitive methods are fast but depend on capacitor behavior and switching arrangement. Electronic loads improve control but introduce power dissipation and thermal issues. Thus, there is a need for a compact, low-cost, and accurate curve-tracing method that is convenient for student laboratories and small-scale PV performance analysis. The method should be capable of sweeping the panel operating point over a wide range, recording electrical quantities in real time, and generating reliable $I-V$ and $P-V$ plots.

The problem addressed in this project is therefore formulated as follows: to design and implement a practical PV curve tracer using a SEPIC converter such that duty-cycle variation changes the equivalent resistance seen by the PV module, enabling full-curve acquisition with real-time sensing of voltage, current, and irradiance. The developed system should remain low-cost, safe for low-power operation, and suitable for simulation, PCB implementation, and laboratory demonstration.

3.1.4 Analysis

The analysis of the proposed method is based on the fact that a DC–DC converter can emulate a variable resistance at its input. When a PV module is connected to the input of a converter, the converter does not appear as a fixed load; instead, its equivalent input resistance depends on the converter gain and the load connected at the output. In the IEEE paper on portable PV tracers, this relationship is expressed by

$$R_{pv} = \frac{V_{pv}}{I_{pv}} = \frac{R_o}{M.M}$$

where R_{pv} is the equivalent resistance seen by the PV module, R_o is the output load resistance, and MM is the converter static gain. This relation is central to the analysis because it explains how the converter can be used as a controlled electrical load.

For SEPIC operation in continuous conduction mode, the static gain follows buck–boost type behavior, so the duty cycle directly affects the effective resistance seen by the PV panel. As duty cycle changes, the PV operating point moves along the nonlinear $I-V$ curve. If the equivalent resistance is very small, the operating point moves near the short-circuit region; if the equivalent resistance is very large, the operating point moves near the open-circuit region. Thus, by sweeping duty cycle over a suitable range, many voltage-current points can be collected across the full operating region of the panel. This is the key idea behind the tracing process.

At each duty-cycle step, the panel voltage and current are measured. The corresponding power is then calculated by

$$P = VI.$$

This allows simultaneous generation of the $I-V$ and $P-V$ curves. The analysis also shows why the converter method is preferable: it provides faster and smoother control of operating point compared to discrete resistor switching and avoids the purely dissipative nature of conventional electronic loads. The uploaded PV tracer paper states that converter-based methods offer strong performance in terms of speed, flexibility, and full-curve sweeping capability.

The component design is also part of the analysis. The converter passive elements are chosen so that current ripple and voltage ripple remain within acceptable limits. In practical implementation, switching frequency, inductor values, capacitor values, and load resistance influence the smoothness of tracing, current ripple, voltage ripple, and heating of components. Higher frequency tends to reduce ripple and passive component size, but may increase switching loss and noise. Therefore, the final design represents a tradeoff among ripple, stability, measurement accuracy, and implementation simplicity.

Overall, the analysis confirms that the SEPIC-based approach is technically suitable for PV curve tracing. It offers a controllable input resistance, full operating-range coverage, real-time data collection, and practical compatibility with low-cost hardware. This makes it an effective choice for the present project, especially for low-power educational and laboratory applications.

3.2 Design Method (PO(a))

The design method of the proposed project is based on the application of power electronics, circuit theory, measurement systems, and simulation tools to develop a practical photovoltaic (PV) curve tracer using a SEPIC converter. The core idea is to use the SEPIC converter as a controllable electrical load for the PV module. Unlike a fixed resistor, the converter presents a variable equivalent resistance at its input, and this resistance changes when the duty cycle is varied. As a result, the operating point of the PV module moves along its nonlinear I–V curve, allowing measurement from near short-circuit condition to near open-circuit condition. This operating principle is suitable for PV characterization because it provides faster and smoother sweeping than conventional resistive or capacitive methods.

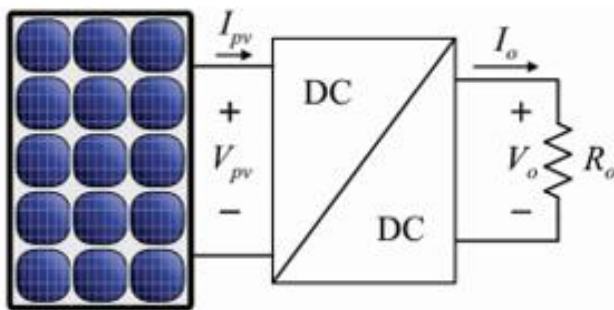


Fig1:PV module with DC-DC load

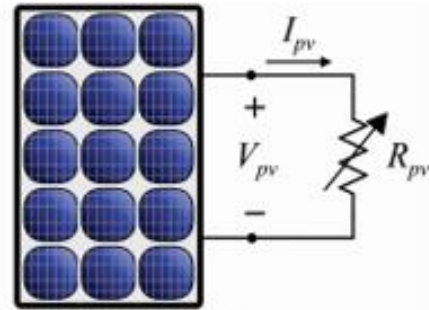


Fig2:Input resistance seen by PV source

R_{PV} depends on value of Duty of a converter,

$$V_o = \frac{D}{1-D} V_{pv}$$

Also Relation between R_{pv} and R_o

$$P_{pv} = P_o$$

$$(V_{pv}/R_{pv})^2 = (V_o/R_o)^2$$

$$R_{pv} = (V_{pv}/V_o)^2 R_o$$

$$R_{pv} = \{(1-D)/D\}^2 R_o$$

To mathematically define the design, the maximum power point resistance of the PV module is first determined as

$$R_{mp} = \frac{V_{mp}}{I_{mp}}$$

where V_{mp} and I_{mp} are the maximum-power voltage and current of the PV module. The converter gain is then related to the output load resistance R_o by

$$M = \sqrt{\frac{R_o}{R_{mp}}}$$

and the corresponding duty cycle is obtained from

$$D = \frac{M}{1+M}$$

These equations are used to determine a suitable operating region for the converter so that the PV panel can be swept over its full curve. In general, the equivalent PV-side resistance can be expressed as

$$R_{pv} = \frac{V_{pv}}{I_{pv}} = \frac{R_o}{M \cdot M}$$

which shows that by changing M , and hence the duty cycle D , the effective input resistance seen by the PV module can be controlled.

The passive elements of the SEPIC converter are selected based on switching frequency and allowable ripple. The inductors are designed to limit current ripple, while the capacitors are chosen to keep voltage ripple within acceptable limits. For the SEPIC converter, the inductor values are estimated using ,

$$L_i, L_o \geq \frac{V_{in} D}{f \Delta I_L}$$

Also Capacitors Values are estimated using,

$$C_1 \geq \frac{V_{in} D}{f R_0 \Delta V_{c_1}} \frac{D}{1-D}$$

$$C_0 \geq \frac{V_{in} D}{f R_0 \Delta V_0} \frac{D}{1-D}$$

Also Value of C_i determine using ,

$$C_i = \frac{4\Delta i_{in}}{\pi^3 f \Delta V_{in}}$$

where V_{in} is the PV-side input voltage, f is the switching frequency, and $\Delta i_{L1}, \Delta i_{L2}$ are the allowable inductor current ripples. Similarly, the capacitors are selected from ripple constraints so that the converter operates smoothly and the measured points remain stable. In this project, practical component values are chosen as $L1=L2=1.3$ mH, $C1=47$ μ F and $C2=10$ μ F, with a switching frequency of 20kHz, which provides a suitable tradeoff between ripple, size, and ease of implementation.

During operation, the Arduino Uno generates the required switching control, and PV voltage and current are measured in real time through the sensing circuit. The output power is then calculated from

$$P=VI$$

and the measured voltage-current pairs are used to construct the I–V curve, while the corresponding power values are used to generate the P–V curve. MATLAB/Simulink is first used to verify the converter behavior and tracing method before hardware fabrication, which reduces design error and allows validation of the chosen converter parameters. After simulation, the circuit is implemented on a custom PCB to improve reliability, reduce loose wiring, and achieve a more compact prototype. The overall design therefore combines mathematical modeling, converter design, sensing, simulation, and hardware realization into a complete low-cost PV characterization system suitable for educational and laboratory use.

3.3 Circuit Diagram

Full Circuit Diagram:

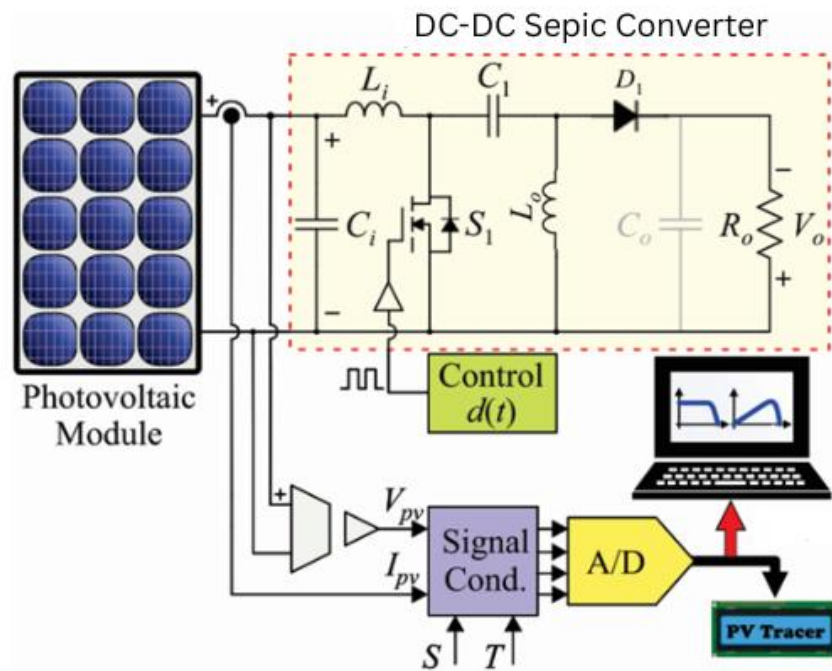


Fig3: Full Circuit Diagram

Simulation Model

3.4 Our Designed Circuit also Simulate at first in Matlab,

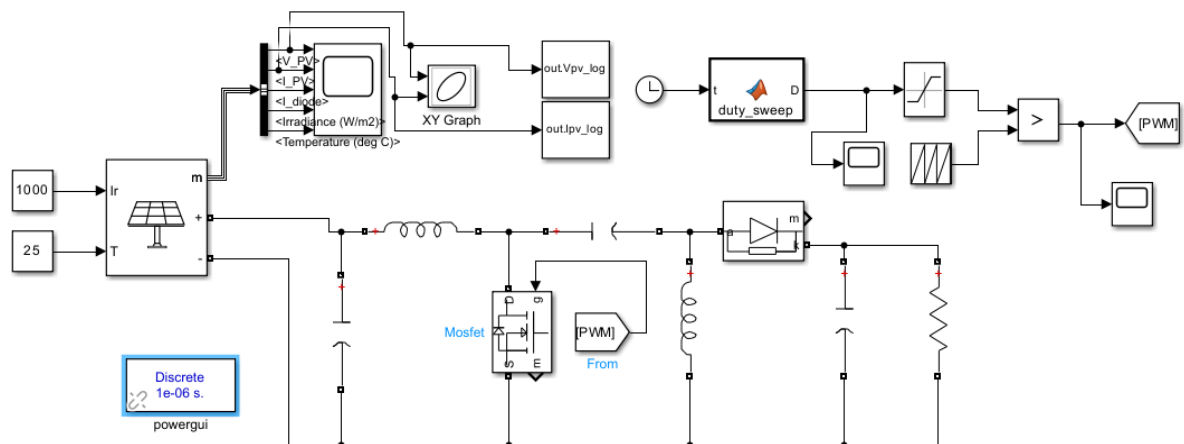


Fig4:Full Circuit diagram in Simulink

3.5 CAD/Hardware Design

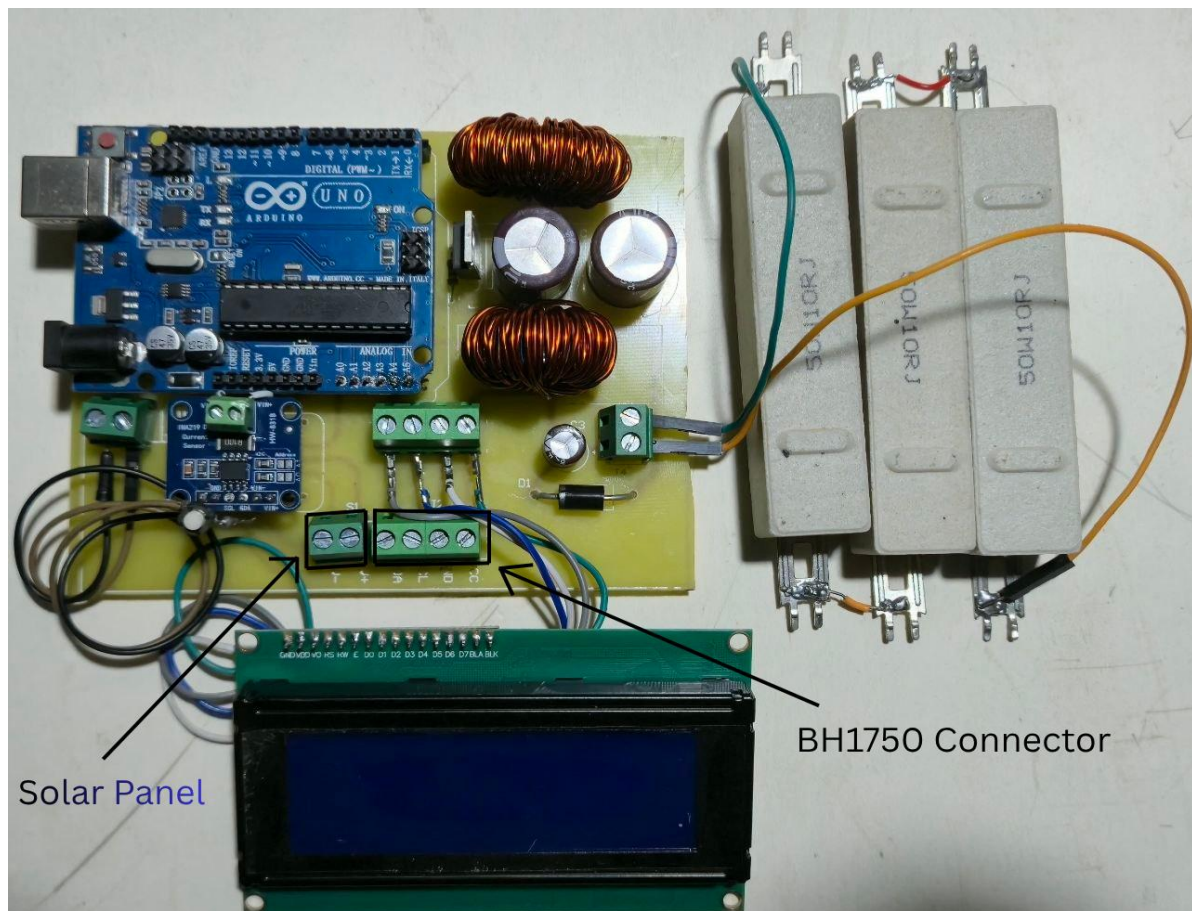


Fig5:Hardware Setup

3.6 PCB Design

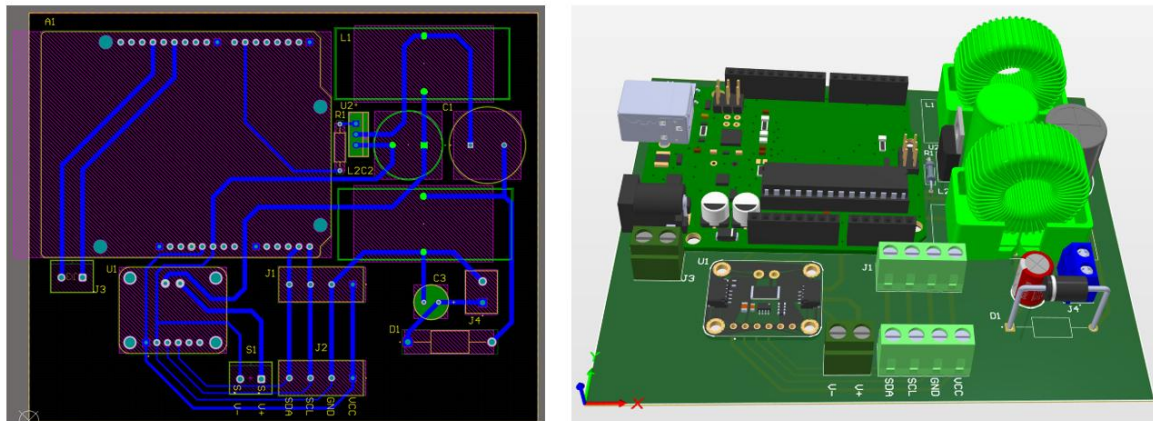


Fig6:2D and 3D view of designed PCB

4 Implementation

4.1 Description

The project titled “**Design of a Solar Curve Tracer Using SEPIC Converter**” is aimed at developing a compact and low-cost system for measuring the electrical characteristics of small photovoltaic (PV) panels. The main purpose of the project is to obtain the **current–voltage (I–V)** and **power–voltage (P–V)** curves of a solar panel under real operating conditions. These curves are important because they provide useful information about the panel’s open-circuit voltage, short-circuit current, maximum power point, and overall performance. Since PV panel output changes with irradiance, temperature, and load condition, practical tracing of these curves is necessary for proper performance analysis.

In this project, a **SEPIC converter** is used as the main curve-tracing element. The converter acts as a controllable electrical load for the PV module. By varying the **duty cycle** of the converter switch, the equivalent resistance seen by the panel changes, which causes the operating point of the PV module to move along its nonlinear I–V characteristic. In this way, multiple voltage-current data points are collected from near short-circuit condition to near open-circuit condition. These points are then used to construct the I–V curve, and the corresponding power values are calculated using

$$P=VI$$

to generate the P–V curve.

Before hardware implementation, the system is first verified in **MATLAB/Simulink** in order to study the converter operation, confirm the tracing principle, and reduce design uncertainty. After simulation, the hardware prototype is developed using a **custom PCB** for better reliability and compactness. The practical system uses an **Arduino Uno** as the main control and data acquisition unit. Voltage and current are measured using the **INA219 sensor**, while

the **BH1750 sensor** is used to estimate light intensity. The measured data are processed and later used to generate the PV characteristics.

The implemented prototype is designed mainly for **low-power PV panels**, specifically panels of **1.5 W and 15 W**, in order to keep the system safe, affordable, and suitable for laboratory testing. The sensing and converter sections are designed within practical voltage and current limits, and the PCB implementation helps avoid loose wiring and improves overall stability of the circuit. During testing, the system successfully records voltage, current, and light intensity data while sweeping the converter duty cycle, and the obtained data produce the expected I–V and P–V curves of the solar panels.

Overall, the implemented project demonstrates that a **SEPIC-converter-based PV curve tracer** can serve as an effective and educationally useful tool for solar panel characterization. The design combines power electronics, sensing, control, simulation, and practical hardware realization in a single platform. It provides a low-cost and hardware-friendly solution for understanding real PV behavior and can be extended in the future for higher-power modules, improved accuracy, and more advanced monitoring features.

5 Design Analysis and Evaluation

5.1 Novelty

The novelty of this project does not lie in proposing a completely new photovoltaic curve-tracing topology. Instead, its contribution is the practical realization of a low-cost, portable, SEPIC-converter-based solar curve tracer for low-power photovoltaic modules. The work was not limited to theoretical study; rather, the complete system was first verified in MATLAB/Simulink and then implemented in hardware, which improved confidence in the design before fabrication. In addition, a custom PCB was designed to make the system more compact, organized, and reliable than a temporary breadboard implementation. The final prototype was also integrated into a portable enclosure, making it more suitable for demonstration and repeated educational use. Another important feature of this work is that it combines Arduino-based duty-cycle control, INA219-based electrical measurement, and BH1750-based light sensing in a single platform for automated $I - V$ and $P - V$ curve tracing. The project also includes practical calibration of the BH1750 sensor, where a setup-specific conversion of Lux/90 was found to provide better irradiance estimation than the commonly cited Lux/122 relation for this setup.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to public health and safety

Public health and safety were considered during both the design and implementation stages. The system was first verified in MATLAB/Simulink before hardware fabrication, which reduced unsafe trial-and-error experimentation. The prototype was intentionally designed for low-power photovoltaic modules, making it safer for laboratory use and student demonstration. The sensing section operates within a limited range of approximately 30 V and 2 A, which helps keep the operating condition controlled and reduces electrical risk. Furthermore, the use of a custom PCB improves reliability by reducing loose wiring and accidental short connections that are common in breadboard setups. The enclosure also

provides physical protection and reduces accidental contact with energized components. In addition, a push-button-controlled operation was used so that the duty-cycle sweep starts only when intentionally triggered by the user.

5.2.2 Considerations to environment

Environmental considerations are relevant because the project supports renewable-energy understanding and solar panel performance evaluation. Since photovoltaic systems are increasingly important for sustainable power generation, a tool that helps characterize their actual behavior contributes positively to renewable-energy engineering practice. The use of simulation before hardware realization reduced unnecessary component wastage and avoided repeated trial-and-error hardware redesign. The PCB-based implementation also improves durability and allows the prototype to be reused for future academic work. Because the system is designed for low-power module testing, its material and energy requirement is much lower than that of larger commercial testing equipment.

5.2.3 Considerations to cultural and societal needs

The project addresses cultural and societal needs by using low-cost and widely available components, making the system suitable for student laboratories and local educational environments. In many developing regions, including Bangladesh, small-scale solar energy systems are commonly used in both rural and urban settings. Therefore, a simple and affordable curve tracer that helps users understand the real performance of a solar panel has practical educational and social value. The portable design also makes the system suitable for demonstration, training, and awareness activities related to renewable energy. Finally, the project gives students practical exposure to simulation, hardware design, sensing, calibration, and data analysis in one complete engineering workflow.

5.3 Investigations (PO(d))

5.3.1 Design of Experiment

The investigation was carried out in two stages. First, the SEPIC-based curve tracer concept was modeled and verified in MATLAB/Simulink to study the expected converter behavior and confirm that duty-cycle variation could shift the operating point of the photovoltaic module over its full operating range. After this verification, the hardware system was developed using an Arduino Uno, INA219 sensor, BH1750 sensor, a SEPIC converter stage, and a custom PCB. During experimental testing, the Arduino varied the PWM duty cycle gradually so that the SEPIC converter behaved as a variable electrical load. At each duty-cycle step, the voltage and current of the PV module were measured, and power was calculated. Lux values were also measured by the BH1750 sensor and later converted into estimated irradiance. A separate calibration experiment was also performed by comparing BH1750 readings against an irradiance meter to improve the irradiance estimation method.

5.3.2 Data Collection

The main data collected during the investigation were photovoltaic module voltage, current, power, and lux during the duty-cycle sweep. These data points were then processed in MATLAB to generate

the $I - V$ and $P - V$ characteristics of the solar panel. From the measured curves, important performance parameters such as V_{oc} , I_{sc} , V_{mpp} , I_{mpp} , and P_{mpp} could be identified. Lux values from the BH1750 sensor were also recorded and used to estimate irradiance after calibration. Practical observations related to system constraints, including the sensor operating limit and the influence of measurement calibration, were also documented.

5.3.3 Results and Analysis

The implemented prototype was tested with low-power photovoltaic modules and was able to successfully perform the intended curve-tracing operation. By sweeping the duty cycle of the SEPIC converter, the system collected real-time voltage and current data from the PV module and generated the corresponding $I-V$ and $P-V$ characteristics. The obtained $I-V$ curves showed the expected nonlinear behavior of a solar panel, while the $P-V$ curves clearly indicated the maximum power region. This confirms that the proposed SEPIC-based approach is suitable for practical PV characterization of small solar panels.

We used 2types solar panel. In different condition,
For 15W solar panel,

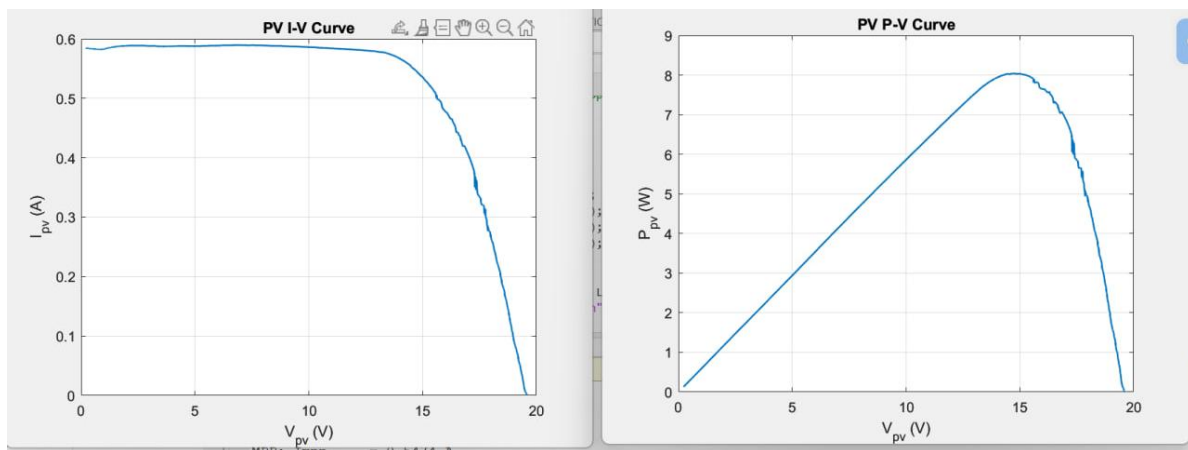


Fig7: I-V and P-V Characteristics of Solar panel(15W panel)

Result:

```
SUMMARY
Voc = 19.5948
Isc = 0.5897
Pmax = 8.0405
Vm = 14.6872
Im = 0.5475
```

Irradiance=600W/m²

Simulation Result:

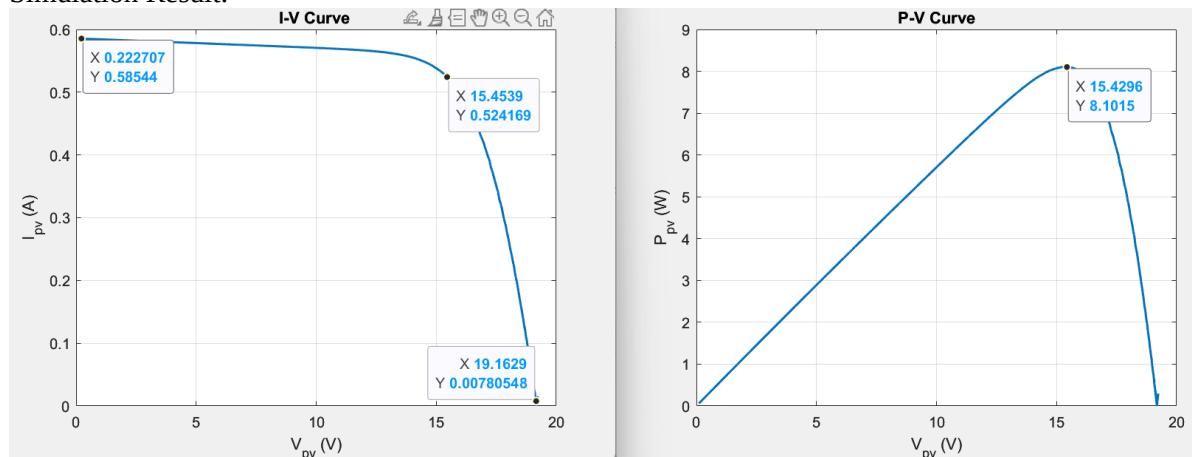


Fig8: Using same Irradiance output from Simulink

$V_{oc} = 19.16$ V
 $I_{sc} = 0.59$ A
 $P_{max} = 8.1015$ W
 $V_{max} = 15.43$ V
 $I_{max} = 0.53$ A

For 1.5W solar panel,

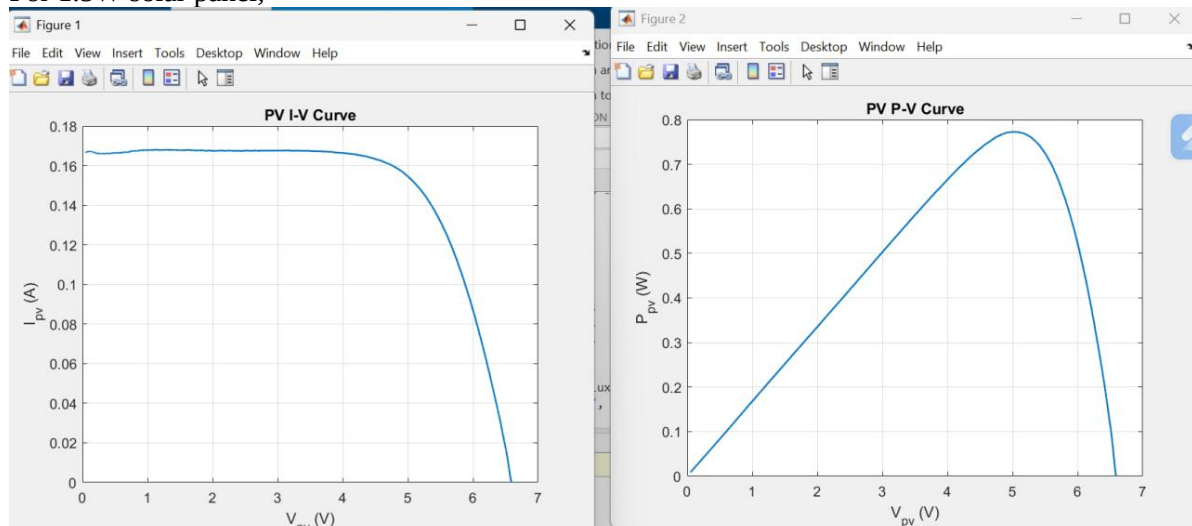


Fig9: I-V and P-V Characteristics of Solar panel(1.5W panel)

```

SUMMARY
Voc = 6.5920
Isc = 0.1681
Pmax = 0.7728
Vm = 5.0348
Im = 0.1535
Avg Lux = 54607.94
DONE
  
```

Irradiance = 610 W/m^2

Simulation Result:

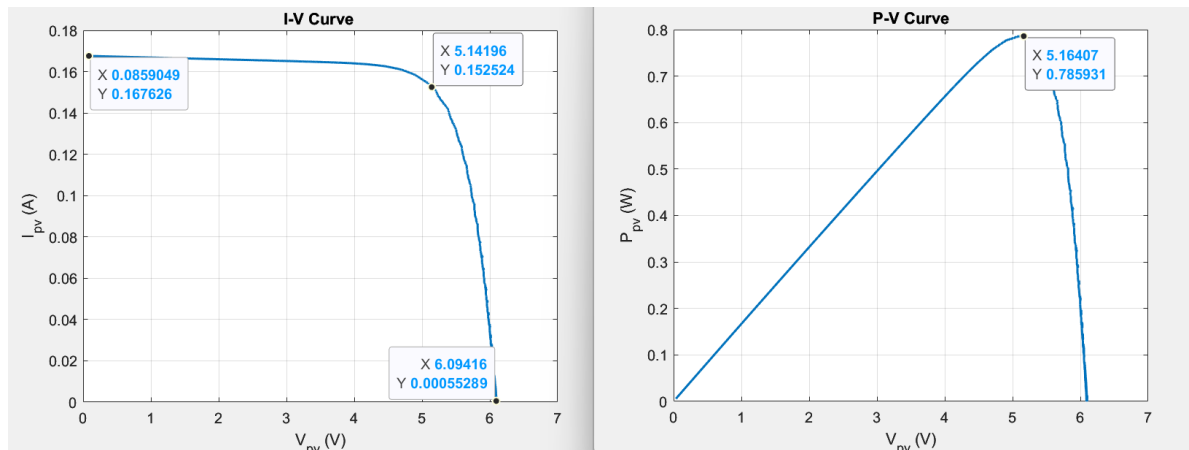


Fig10: Using same Irradiance output from Simulink

$$V_{oc}=6.09V$$

$$I_{sc}=0.17A$$

$$P_{max}=0.786W$$

$$V_{max}=5.164V$$

$$I_{max}=0.15A$$

Efficiency and Fill Factor Calculation:

For 15W Solar Panel:

$$\text{Area, } A = 38 \text{ cm} \times 35 \text{ cm} = 1330 \text{ cm}^2 = 0.133 \text{ m}^2$$

$$\text{Irradiance, } Irr = 600 \text{ W/m}^2$$

$$P_{max} = 8.0405 \text{ W}$$

$$\text{Efficiency, } \eta = P_{max} / (A \times Irr)$$

$$= (8.0405 / (0.133 \times 600))$$

$$= 0.101$$

$$= 10.1\%$$

$$V_{oc} = 19.6 \text{ V}$$

$$I_{sc} = 0.59 \text{ A}$$

$$\text{Fill Factor, } FF = P_{max} / (V_{oc} \times I_{sc})$$

$$= (8.0405 / (19.6 \times 0.59))$$

$$= 0.695$$

For 1.5W Solar Panel:

$$\text{Area, } A = 16.5 \text{ cm} \times 6.5 \text{ cm} = 107.25 \text{ cm}^2 = 0.010725 \text{ m}^2$$

$$\text{Irradiance, } Irr = 610 \text{ W/m}^2$$

$$P_{max} = 0.7728 \text{ W}$$

$$\text{Efficiency, } \eta = P_{max} / (A \times Irr)$$

$$= (0.7728 / (0.010725 \times 610))$$

$$= 0.1181$$

$$= 11.81\%$$

$$V_{oc} = 6.59 \text{ V}$$

$$I_{sc} = 0.1681 \text{ A}$$

$$\text{Fill Factor, } FF = P_{max} / (V_{oc} \times I_{sc})$$

$$= (0.7728 / (6.59 \times 0.1681))$$

$$= 0.6976$$

Analysis:

The experimental results confirm that the proposed SEPIC-based solar curve tracer was able to successfully obtain the I–V and P–V characteristics of both the 15 W and 1.5 W photovoltaic panels. In both cases, the overall curve shapes followed the expected photovoltaic behavior, and the measured maximum power points were close to the simulated values. This indicates that the proposed design, simulation model, and hardware implementation are fundamentally correct. However, a small difference between simulation and experimental results was observed, which is normal in practical power-electronic systems. The simulation was carried out under ideal or simplified assumptions, whereas the hardware prototype operated under real conditions where several non-ideal factors were present. These include switching ripple in the SEPIC converter, component tolerance of inductors and capacitors, voltage drop across the MOSFET and diode, sensor inaccuracy, PCB parasitic effects, measurement noise, and slight variation in irradiance during testing. In addition, the irradiance estimation using the BH1750 sensor required practical calibration, which also contributed to deviation from the ideal simulated condition. Therefore, the difference between simulation and experiment does not indicate failure; rather, it shows the expected gap between an ideal model and a real hardware implementation. Overall, the experimental results demonstrate that the prototype is capable of practical low-power PV characterization with acceptable accuracy for laboratory and educational use.

5.3.4 Interpretation and Conclusions on Data

The collected data indicate that the proposed SEPIC-based tracer is a workable and practical solution for low-cost solar panel characterization. The system can trace the electrical behavior of a PV module by sweeping the converter duty cycle and measuring voltage and current at multiple operating points. The MATLAB-generated curves provide a useful basis for performance analysis and parameter extraction. The results also show that although the BH1750 sensor can be used for irradiance estimation, accurate use requires calibration under the actual experimental condition. Overall, the investigation supports the conclusion that the developed system is suitable for educational and low-power experimental use, though not yet intended as a replacement for professional commercial instrumentation.

5.4 Limitations of Tools (PO(e))

Several limitations of the tools used in this project were identified. MATLAB/Simulink is highly useful for theoretical verification, but simulation cannot fully model all practical non-idealities such as switching noise, wiring resistance, sensor offset, sunlight fluctuation, and component tolerance. The INA219 sensor has a safe operating range of about 30 V and 2 A, which restricts the present design to low-power photovoltaic modules. The BH1750 sensor measures lux rather than irradiance directly, so it cannot provide a physically exact irradiance value without calibration, and even after calibration it should still be treated as an estimate. The original Lux/122 conversion was not sufficiently accurate for this setup, demonstrating that optical measurement conditions affect the result. In addition, Arduino Uno has limited speed, memory, and processing capability compared to dedicated data-acquisition systems, which can limit sweep rate and data resolution. Finally, while PCB implementation improves reliability and portability, measurement accuracy can still be affected by layout quality, soldering condition, and the tolerance of passive components used in the converter.

5.5 Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

The project provides a low-cost platform for solar panel performance testing, which is useful in academic laboratories and student learning environments. By showing the behavior of $I - V$ and $P - V$ a solar panel, it helps users understand that real panel performance may differ from datasheet values due to environmental condition, aging, or degradation. Since the system is based on low-cost components and a custom PCB, it is practical for local educational purposes and can help improve understanding of renewable-energy systems in contexts where professional instruments are not easily available.

5.5.2 Assessment of Health and Safety Issues

The design includes several features that improve safe operation. Simulation verification reduced uncertainty before hardware construction. The voltage and current limits of the measurement system help keep the prototype within a controlled operating range. The PCB-based hardware reduces exposed loose connections, and the enclosed design lowers the chance of accidental contact. However, safe operation still depends on proper grounding, insulation, and using the system only within its intended electrical limits. Therefore, while the project is relatively safe for laboratory-scale use, it still requires responsible handling during operation.

5.6 Sustainability Evaluation (PO(g))

This project supports sustainability by promoting renewable-energy learning and practical photovoltaic testing. The use of MATLAB/Simulink before fabrication reduced unnecessary hardware waste and improved component utilization. The custom PCB and enclosure make the system more durable and reusable for repeated academic use, which reduces repeated rebuilding. The low-power nature of the test setup also means that material and energy consumption remain limited. However, sustainability also requires proper handling and disposal of damaged electronic components and batteries, if any are used in future versions. Thus, while the project makes a positive contribution to renewable-energy education, responsible lifecycle handling of electronics remains important.

5.7 Ethical Issues (PO(h))

Ethical engineering practice in this project mainly involves honest reporting, responsible operation, and proper acknowledgment of sources and tools. The measured results must be presented truthfully, including the limitations of the sensors and the calibration process. Since the BH1750 sensor does not measure irradiance directly, its output must be reported as a calibrated estimate rather than an exact or certified irradiance value. It is also ethically important not to manipulate data in order to force agreement with theory or datasheet curves; differences between experiment and expectation should instead be explained transparently. The system should only be used within its safe operating range of approximately 30 V and 2 A, because operation beyond this range would be technically irresponsible and potentially unsafe. In addition, all reference papers, software tools, calibration methods, and external

resources used in the project must be properly cited and acknowledged. Overall, ethical responsibility in this work means presenting the prototype honestly as an educational and experimental instrument, not overstating its precision or commercial readiness.

6 Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

2006074	Literature review, Simulink verification, testing, calibration.
2006077	Literature review, circuit design, Arduino coding, sensor integration, testing, calibration.
2006096	Literature review, Simulink verification, circuit design, hardware implementation, testing, PCB design.
2006138	Literature review, Simulink verification, Arduino coding, testing, MATLAB analysis

6.2 Mode of TeamWork

The team adopted a modular development strategy, dividing tasks into sensing, communication, actuation, and testing. Weekly check-ins were scheduled to ensure synchronization. Team members worked in pairs during lab hours and communicated progress through online meetings and group chat. Tasks were assigned based on individual strengths in hardware, firmware, and system integration.

6.3 Diversity Statement of Team

The project team consisted of members with diverse skills, backgrounds, and areas of expertise, which contributed significantly to the successful completion of the project. Each member brought unique strengths in areas such as circuit design, simulation, programming, hardware implementation, and data analysis. This diversity allowed the team to approach problems from multiple perspectives and develop well-balanced solutions.

The team maintained an inclusive and collaborative environment where all members were encouraged to share ideas, provide feedback, and participate in decision-making. Tasks were distributed based on individual strengths while ensuring that everyone remained involved in the overall development process. Regular discussions and knowledge sharing helped improve collective understanding and ensured effective coordination among team members.

This diversity in skills and teamwork enhanced the quality of the project by combining theoretical knowledge with practical implementation, ultimately leading to a more efficient, reliable, and well-integrated system.

6.4 Log Book of Project Implementation

Date	Task/Phase	Description	Assigned to	Status	Req. Time	Action Taken	Impact
30/11/2025 – 04/01/2026	Topic Selection and Literature Review	Studied existing PV curve-tracing methods and selected a SEPIC-converter-based approach for a low-cost hardware implementation.	All members	Completed	5 weeks	Reviewed papers, compared tracing methods, and finalized project scope.	Research
13/01/2026 – 14/02/2026	Simulink Modeling and Verification	Developed the SEPIC-based PV tracer model in MATLAB/Simulink and verified I–V tracing behavior before hardware implementation.	2006074, 2006096, 2006138	Completed	1 month	Validated converter operation, duty-cycle sweep, and expected I–V/P–V response.	Simulation
27/02/2026 – 05/03/2026	Circuit Design and Component Selection	Selected practical values of inductors, capacitors, MOSFET, sensors, and switching frequency for the prototype.	2006077, 2006096	Completed	1 week	Designed the main circuit and finalized hardware component list.	Circuit Design
05/03/2026 – 26/03/2026	Arduino Coding and Sensor Integration	Implemented Arduino code for duty-cycle control, INA219 voltage/current sensing, and BH1750 lux measurement.	2006077, 2006138	Completed	3 weeks	Integrated sensors with controller and verified real-time data collection.	Firmware
05/03/2026 – 26/03/2026	Hardware Assembly and Testing	Assembled the converter and sensing circuit, connected PV panels, and tested prototype operation under practical conditions.	2006096, 2006138	Completed	3 weeks	Checked wiring, switching operation, measurement stability, and panel response.	Testing
30/03/2026 – 01/04/2026	PCB Design	Designed a custom PCB layout to improve reliability, compactness, and ease of implementation.	2006096	Completed	3 days	Prepared PCB layout and verified component placement and routing.	Implementation
04/04/2026	BH1750 Calibration with Irradiance Meter	Calibrated the BH1750 sensor against an irradiance meter and observed that Lux/90 gave better estimation than Lux/122 in the experimental setup.	2006074, 2006077	Completed	1 day	Performed calibration test and updated irradiance conversion method.	Calibration
02/04/2026 – 05/04/2026	Final Data Analysis and Presentation	Processed experimental data, generated I–V and P–V curves in MATLAB, and prepared the final report and presentation.	All members	Completed	4 days	Compared simulation and hardware results, summarized findings, and finalized presentation materials.	Analysis

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

A team of engineering students has developed a low-cost device to better understand how solar panels perform in real-world conditions. Using a smart electronic circuit called a SEPIC converter, the system quickly measures how much power a solar panel can produce at different conditions. The device automatically generates performance curves, helping identify the maximum usable power from the panel. Tested on small solar panels, the system showed reliable and accurate results. This innovation offers an affordable and practical solution for solar testing, making it useful for education, research, and small-scale renewable energy applications, especially in regions where access to advanced testing equipment is limited.

7.2 User Manual

7.2.1 Setup Steps:

1. Connect the solar panel terminals to the input of the SEPIC converter.
2. Connect the BH1750 light sensor to the Arduino for irradiance measurement.
3. Ensure all components are properly connected on the PCB or breadboard.
4. Connect the Arduino to a computer for data monitoring (optional).
5. Power up the system using a suitable DC supply or the PV panel itself.
6. Place the solar panel under sunlight or a controlled light source.
7. Press the Push Button.

7.2.2 Usage:

- The system automatically sweeps the duty cycle of the SEPIC converter.
- Voltage and current data are collected in real time.
- I–V and P–V characteristics are generated from the measured data.
- Maximum power point (MPP) can be identified from the P–V curve.
- Data can be visualized using MATLAB or serial monitor output.
- Ensure stable lighting conditions for accurate measurement.

7.3 YouTube Link

Link: <https://youtu.be/cL9pOd7cLNM>

8 Project Management and Cost Analysis (PO(k))

8.1 Bill of Materials

Components	amount	Per unit Cost(tk)	Total Cost
Arduino uno	1	750	750
INA219	1	214	214
LCD display(20x4)	1	420	420
I2C adapter	1	70	70
Inductor Core	2	40	80
Copper Wire	60ft.	500	500
47 μ F Capacitor	1	26	26
10 μ F Capacitor	1	7	7
1000 μ F Capacitor	1	40	40
Diode	1	8	8
Mosfet	1	80	80
Power Resistor	3	210	630
BH1750	1	250	250
Solar Panel(1.5W)	1	450	450
Push Button	1	5	5
Screw Connector	5	15	75
wire	N/A	60	60
PCB Board	1	190	190
Board Print	1	100	100
Male Header	30	2	60
Jumper Wire	N/A	80	80
Total			4095

8.2 Timeline of Project Implementation

Tasks	30 Nov– 04 Jan	13 Jan–14 Feb	27 Feb– 05 Mar	05–26 Mar	30 Mar– 01 Apr	04 Apr	02–05 Apr
Topic Selection and Literature Review	■						
Simulink Modeling and Verification		■					
Circuit Design and Component Selection			■				

Arduino Coding and Sensor Integration				■			
Hardware Assembly and Testing				■			
PCB Design					■		
BH1750 Calibration with Irradiance Meter						■	
Final Data Analysis and Presentation							■

9 Future Work (PO(I))

The present project demonstrates a low-cost and portable photovoltaic curve tracer for small solar panels using a SEPIC converter. However, there are several directions in which this work can be improved and extended in the future.

First, the system can be upgraded to support higher-power photovoltaic modules. The current design is mainly intended for low-power panels because the sensing and converter stages are limited by the ratings of the present components. In future work, higher-rated current and voltage sensors, more robust switching devices, and appropriately designed converter components can be used so that the system can safely test larger solar panels. This would make the tracer more versatile and suitable for a broader range of practical photovoltaic applications.

Second, temperature sensing can be added to the system. In the present work, irradiance estimation is included, but photovoltaic performance is strongly affected by both irradiance and temperature. Since parameters such as open-circuit voltage, maximum power, and overall efficiency vary with temperature, adding a temperature sensor would allow a more complete analysis of the solar panel behavior. This would also make it possible to compare measured results more accurately with datasheet values under different operating conditions.

Third, the measurement accuracy of the prototype can be improved. Although the current system is sufficient for educational and experimental use, further work can focus on reducing measurement error through better signal filtering, improved shielding, and more reliable protection circuitry. In addition, more careful calibration of the sensors can improve the quality of the measured voltage, current, and irradiance-related data. Such improvements would increase confidence in the extracted $I - V$ and $P - V$ characteristics and make the device more dependable for detailed performance evaluation.

Fourth, a more advanced version of the tracer can be developed with automatic data storage and improved user interface. At present, the system mainly focuses on obtaining and displaying the curve data. In future work, data logging features can be added so that measured values are automatically stored for later analysis. Wireless communication or a dedicated PC/mobile interface can also be integrated to allow easier visualization, monitoring, and transfer of measurement results. This would make the system more user-friendly and more suitable for repeated field or laboratory testing.

Fifth, the project can be extended from basic curve tracing to fault detection and condition

monitoring of photovoltaic modules. The $I - V$ and $P - V$ curve contains useful information about panel degradation, shading effects, increased series resistance, reduced shunt resistance, and other abnormalities. In future work, software-based analysis can be added to automatically identify deviations in the measured curve and relate them to possible faults. This would transform the system from a simple measurement device into a more intelligent diagnostic tool for solar panel health assessment.

Finally, the present prototype can serve as a foundation for further learning and development in renewable-energy instrumentation. The work already combines power electronics, sensing, embedded control, PCB implementation, and data analysis in one system. Future improvements can therefore build on this base to produce a more accurate, more robust, and more practical instrument. In this way, the project has strong potential not only as an academic exercise, but also as a starting point for more advanced research and development in photovoltaic testing systems.

10 References

- M. Veerachary, T. Senjyu, and K. Uezato, “Design of a portable photovoltaic I–V curve tracer based on a DC–DC converter,” *IEEE Transactions on Energy Conversion*, vol. 19, no. 2, pp. 475–481, Jun. 2004.
- M. A. Rahman, M. S. Islam, and M. R. Islam, “I–V curve tracer development for the performance analysis of a photovoltaic system,” *International Journal of Engineering Research & Technology (IJERT)*, vol. 11, no. 8, pp. 1246–1250, Aug. 2022.

Appendix

Full Source Code of Firmware

```
#include <Wire.h>
#include <Adafruit_INA219.h>
#include <BH1750.h>
#include <LiquidCrystal_I2C.h>

Adafruit_INA219 ina219;
BH1750 lightMeter;
LiquidCrystal_I2C lcd(0x27, 20, 4); // change to
0x3F if needed

// ===== SETTINGS
=====
// PWM (Timer1 -> Pin 9)
const uint16_t TOP = 799; // 16MHz/(TOP+1)=20kHz
with prescaler=1
const uint8_t PWM_PIN = 9; // OC1A

// Button
const uint8_t BTN_PIN = 2;
const uint16_t DEBOUNCE_MS = 40;

// Sweep
const float D_MIN = 0.00f;
const float D_MAX = 0.99f;
const uint16_t NPTS = 1000;

const uint16_t SETTLE_MS = 3;
const uint8_t AVG_SAMPLES = 10;
const uint16_t SAMPLE_DELAY_MS = 1;

// LCD update interval
const uint16_t LCD_UPDATE_EVERY = 20;

// Lux to Irradiance conversion
const float LUX_TO_IRR_DIV = 90.0f;

bool sweeping = false;

// ===== HELPERS
=====
static inline uint16_t dutyToOCR1A(float duty) {
    if (duty < 0.0f) duty = 0.0f;
    if (duty > 1.0f) duty = 1.0f;
    uint32_t val = (uint32_t)((duty * (TOP + 1)) +
0.5f);
    if (val > TOP) val = TOP;
    return (uint16_t)val;
}

void clearLine(uint8_t row) {
    lcd.setCursor(0, row);
    lcd.print(" ");
}

void setupTimer1_20kHz() {
    pinMode(PWM_PIN, OUTPUT);

    TCCR1A = 0;
    TCCR1B = 0;
    TCNT1 = 0;

    // Fast PWM mode 14 (TOP=ICR1)
    TCCR1A |= (1 << COM1A1);
    TCCR1A |= (1 << WGM11);
    TCCR1B |= (1 << WGM13) | (1 << WGM12);

    ICR1 = TOP;
    TCCR1B |= (1 << CS10);

    OCR1A = 0;
}

void setupSensors() {
    Wire.begin();

    // INA219
    ina219.begin();
    ina219.setCalibration_32V_2A();

    // BH1750
    lightMeter.begin(BH1750::CONTINUOUS_HIGH_RES_MODE);

    delay(200);
}

void setupLCD() {
    lcd.init();
    lcd.backlight();

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("PV I-V Tracer");
    lcd.setCursor(0, 1);
    lcd.print("INA219 + BH1750");
    lcd.setCursor(0, 2);
    lcd.print("20x4 LCD Ready");
    lcd.setCursor(0, 3);
    lcd.print("Press Button...");
}

bool buttonPressedEvent() {
    static bool lastStableState = HIGH;
    static bool lastReadState = HIGH;
    static unsigned long lastChangeTime = 0;

    bool readState = digitalRead(BTN_PIN);

    if (readState != lastReadState) {
        lastChangeTime = millis();
        lastReadState = readState;
    }

    if ((millis() - lastChangeTime) > DEBOUNCE_MS) {
        if (readState != lastStableState) {
            lastStableState = readState;
            if (lastStableState == LOW) return true;
        }
        return false;
    }
}

// ===== LCD FUNCTIONS
=====
void showRunningScreen(uint16_t i, float Vpv, float
Ipv, float P, float irradiance) {
    lcd.clear();

    lcd.setCursor(0, 0);
    lcd.print("Sweep Running...");

    clearLine(1);
    lcd.setCursor(0, 1);
    lcd.print("Pt:");
    lcd.print(i);
    lcd.print("/");
    lcd.print(NPTS - 1);

    clearLine(2);
    lcd.setCursor(0, 2);
    lcd.print("V:");
    lcd.print(Vpv, 2);
    lcd.setCursor(10, 2);
    lcd.print("I:");
    lcd.print(Ipv, 2);

    clearLine(3);
    lcd.setCursor(0, 3);
    lcd.print("P:");
```

```

    lcd.print(P, 2);
    lcd.setCursor(10, 3);
    lcd.print("Ir:");
    lcd.print(irradiance, 1);
}
void showFinalScreen(float Voc, float Isc, float Vm,
float Im, float Pmax, float irrAvg) {
    lcd.clear();

    clearLine(0);
    lcd.setCursor(0, 0);
    lcd.print("Voc:");
    lcd.print(Voc, 2);
    lcd.setCursor(10, 0);
    lcd.print("Isc:");
    lcd.print(Isc, 2);

    clearLine(1);
    lcd.setCursor(0, 1);
    lcd.print("Vm:");
    lcd.print(Vm, 2);
    lcd.setCursor(10, 1);
    lcd.print("Im:");
    lcd.print(Im, 2);

    clearLine(2);
    lcd.setCursor(0, 2);
    lcd.print("Pmax:");
    lcd.print(Pmax, 2);
    lcd.print(" W");
    lcd.setCursor(10, 2);
    lcd.print("Ir:");
    lcd.print(irrAvg, 2);

    clearLine(3);
    lcd.setCursor(0, 3);
    lcd.print("Enter Push Buttotn");
}

// ===== SWEEP FUNCTION
=====
void doSweepOnce() {
    sweeping = true;

    OCR1A = dutyToOCR1A(D_MIN);
    delay(200);

    // Extracted final values
    float Voc = 0.0f;
    float Isc = 0.0f;
    float Pmax = -1.0f;
    float Vm = 0.0f;
    float Im = 0.0f;

    // Average irradiance during full sweep
    double irrSum = 0.0;
    uint16_t irrCount = 0;
    Serial.println("idx duty Vpv_V Ipv_A P_W
Irradiance_Wm2");

    for (uint16_t i = 0; i < NPTS; i++) {
        float duty = D_MIN + (D_MAX - D_MIN) * ((float)i
/ (float)(NPTS - 1));

        OCR1A = dutyToOCR1A(duty);
        delay(SETTLE_MS);

        // ---- Average INA219 readings ----
        double vSum = 0.0;
        double iSum = 0.0;

        for (uint8_t k = 0; k < AVG_SAMPLES; k++) {
            float v = ina219.getBusVoltage_V();
            float ia = ina219.getCurrent_mA() / 1000.0f;
            vSum += v;
            iSum += ia;
            delay(SAMPLE_DELAY_MS);
        }

```

```

float Vpv = (float)(vSum / AVG_SAMPLES);
float Ipv = (float)(iSum / AVG_SAMPLES);
if (Ipv < 0) Ipv = -Ipv;

float P = Vpv * Ipv;

// ---- BH1750 lux -> irradiance ----
float lux = lightMeter.readLightLevel();
if (lux < 0) lux = 0.0f;

float irradiance = lux / LUX_TO_IRR_DIV;

irrSum += irradiance;
irrCount++;

// ---- Extract final values ----
if (Vpv > Voc) Voc = Vpv;
if (Ipv > Isc) Isc = Ipv;

if (P > Pmax) {
    Pmax = P;
    Vm = Vpv;
    Im = Ipv;
}

// ---- Serial output ----
Serial.print(i);          Serial.print(" ");
Serial.print(duty, 6);     Serial.print(" ");
Serial.print(Vpv, 6);      Serial.print(" ");
Serial.print(Ipv, 6);      Serial.print(" ");
Serial.print(P, 6);        Serial.print(" ");
Serial.println(irradiance, 3);

// ---- LCD live update ----
if (i % LCD_UPDATE_EVERY == 0) {
    showRunningScreen(i, Vpv, Ipv, P, irradiance);
}
}

OCR1A = 0;
sweeping = false;
float irrAvg = 0.0f;
if (irrCount > 0) {
    irrAvg = (float)(irrSum / irrCount);
}

// ---- Final LCD ----
showFinalScreen(Voc, Isc, Vm, Im, Pmax, irrAvg);

// ---- Final Serial summary ----
Serial.println();
Serial.println("SUMMARY");
Serial.print("Voc = "); Serial.println(Voc, 4);
Serial.print("Isc = "); Serial.println(Isc, 4);
Serial.print("Pmax = "); Serial.println(Pmax, 4);
Serial.print("Vm = "); Serial.println(Vm, 4);
Serial.print("Im = "); Serial.println(Im, 4);
Serial.print("Avg Irradiance = ");
Serial.println(irrAvg, 3);
Serial.println("DONE");
}

// ===== ARDUINO SETUP/LOOP
=====
void setup() {
    Serial.begin(115200);

    pinMode(BTN_PIN, INPUT_PULLUP);

    setupTimer1_20kHz();
    setupSensors();
    setupLCD();

    OCR1A = 0;
}

void loop() {
    if (!sweeping && buttonPressedEvent()) {
        doSweepOnce();
    }
}

```

Matlab Code

```
clc; clear; close all;

filename = "data.txt"; % your file

% Read space-separated numeric data
M = readmatrix(filename, 'Delimiter', ' ');

% Columns: [idx duty V I P Irr_wm2]
idx = M(:,1);
duty = M(:,2);
V = M(:,3);
I = M(:,4);
P = M(:,5);

% If Lux and Irr are present
hasIrr = size(M,2) >= 6;
if hasIrr
    Irr = M(:,6); % W/m^2
else
    Irr = [];
end

% Remove bad rows
if hasIrr
    valid = ~(isnan(V) | isnan(I) | isnan(duty) | isnan(Irr));
else
    valid = ~(isnan(V) | isnan(I) | isnan(duty));
end

idx = idx(valid);
duty = duty(valid);
V = V(valid);
I = I(valid);
P = P(valid);
if hasIrr
    Irr = Irr(valid);
end

% Average irradiance (for the whole sweep)
if hasIrr
    Irr_avg = mean(Irr);
else
    Irr_avg = NaN;
end

% Sort by Voltage for smooth curves
[Vs, k] = sort(V);
Is = I(k);
Ps = P(k);

% --- I-V ---
figure;
plot(Vs, Is, 'LineWidth', 1.2); grid on;
xlabel('V_{pv} (V)');
ylabel('I_{pv} (A)');
if hasIrr
    title(sprintf('PV I-V Curve (Avg Irradiance = %.2f W/m^2,)', Irr_avg));
else
    title('PV I-V Curve');
end

% --- P-V ---
figure;
plot(Vs, Ps, 'LineWidth', 1.2); grid on;
xlabel('V_{pv} (V)');
ylabel('P_{pv} (W)');
if hasIrr
    title(sprintf('PV P-V Curve (Avg Irradiance = %.2f W/m^2,)', Irr_avg));
end

else
    title('PV P-V Curve');
end

% --- Optional: print approximate Voc, MPP ---
Voc = max(Vs);
[Pmpp, km] = max(Ps);
Vmpp = Vs(km);
Impp = Is(km);

fprintf("Voc (approx) = %.4f V\n", Voc);
fprintf("MPP: Pmpp = %.4f W\n", Pmpp);
fprintf("MPP: Vmpp = %.4f V\n", Vmpp);
fprintf("MPP: Impp = %.4f A\n", Impp);

if hasIrr
    fprintf("Avg Irradiance= %.2f W/m^2\n", Irr_avg);
end

lcd.print(P, 2);
lcd.setCursor(10, 3);
lcd.print("Ir:");
```